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Lumina Stream Architecture

This document provides a technical overview of the Lumina Stream application, detailing the configuration steps, API design, and schema definitions.

1. System Requirements

Before initializing the local environment, you must ensure that your system meets the baseline configuration standards.

1.1 Dependencies

The following internal tools must be present on your host machine: * Python 3.10+ * Docker Engine 24.0.0 or higher * Valid TLS certificates in your local keystore

Security Notice

Never commit your `.env` file to version control. All API keys and secrets should be injected at runtime via your CI/CD pipeline variables.

2. Network Configuration

The service routes all incoming traffic through a centralized gateway. You can define specific overrides in your `routing.conf` file using standard proxy directives.

2.1 Example Routing Block

To bypass the standard auth middleware for public assets, append the following rule. Notice the use of the `Cache-Control` header.

```
location /public/ {
    proxy_pass http://cdn_backend;
    proxy_set_header Host $host;
    add_header Cache-Control "public, max-age=31536000";
}
```

3. Data Schema Mappings

When interacting with the ingestion endpoints, payloads are automatically serialized and typed against the database schema.

JSON Primitive	Database Column	Nullable	Index Type
`uuid`	`UUID PRIMARY KEY`	False	B-Tree
`timestamp`	`TIMESTAMP WITH TIME ZONE`	False	Brin
`metadata`	`JSONB`	True	GIN
`is_active`	`BOOLEAN`	False	None

3.1 Field Validations

All text fields are sanitized upon entry. If a payload violates the schema, the API will respond with an HTTP `400 Bad Request` and return a standard error trace.